

Per-Seed Training Results — PPO vs ctrl vs cPPO

Matrix-v2, 3-arm batch · commit 567e4c0, tag matrix-v2 · 10 seeds · 1500 iterations · Isaac-Lift-Cube-UR5e-v0

Scope and provenance

This appendix reports **training-time** telemetry for every individual seed of the matrix-v2 three-arm batch, so that the pooled statistics quoted in the Results chapter can be audited seed by seed. Each arm was trained for 1500 iterations at *num_envs* = 4096 using rsl_rl 3.0.1 against the frozen weld environment. The three arms differ from one another by exactly one variable: **PPO** is the unconstrained baseline; **ctrl** adds the cost critic but pins the Lagrange multiplier to zero, so the constraint cannot influence the policy; **cPPO** releases the multiplier against an episodic cost budget of 25.

All values are read from the per-run TensorBoard exports in *results/tb_csv/* by *results/scripts/make_per_seed_tables.py*. Runs are selected by dated directory path, never by run label, because three superseded pre-audit cPPO runs from 2026-07-30 still carry the same labels as the current ones. All thirty source runs used here were confirmed to be 2026-08-01 runs.

How to read the three columns per arm

final is the value at the last logged iteration (1499). **mean** is the arithmetic mean over all 1500 logged iterations — it includes the untrained early phase, so it describes the area under the learning curve rather than converged performance, and it is systematically pessimistic for reward and optimistic for cost. **tail** is the mean over the final 10% of iterations (150 points). **The tail column is the one to quote:** it is the statistic used throughout *MATRIX_V2_PARTIAL_3ARM.md* and the Results chapter, and it is the only one of the three that will cross-reference against those documents.

Notes on specific rows

† **Mean episode cost, PPO column.** The tag *Loss/mean_episode_cost* is written only by the Lagrangian runner, so the PPO arm never logged it and the value shown is **ctrl's**. This substitution is valid, not a convenience: PPO and ctrl were shown to be bitwise identical — all 68 actor and reward-critic tensors byte-for-byte equal inside the two checkpoints, on all ten seeds, reconfirmed independently at evaluation time. The ctrl arm exists precisely to measure the episodic cost incurred by a policy that does not act on it. Note that PPO and ctrl are consequently identical in *every* row of *every* table below; this was verified numerically across all seeds, metrics and statistics, and is the batch's central validity check rather than a duplication error.

Violation rows. These are soft-margin step fractions — the fraction of control steps on which a continuous safety margin fell the wrong side of a threshold, measured on a still-exploring policy during training. They are **not** the frozen-policy safety figures reported in the Results chapter, which count true singularity crossings ($w < 1e-4$) and per-episode joint-limit contact over 30,000 deterministic evaluation episodes. The two must not be substituted for one another; the soft-margin fraction exaggerates differences by testing a binary threshold on a continuous quantity.

Dispersion convention. Where this appendix's per-seed values are pooled and compared against *MATRIX_V2_PARTIAL_3ARM.md*, note that the standard deviations in that document are population standard deviations (ddof = 0). Sample standard deviations computed from the ten seeds here are larger by a factor of $\sqrt{10/9} \approx 1.054$. Both are correct; state which convention is used.

Scope limit. This is a three-of-five-arm batch. The cppo10/cppo15 binding-budget arm and the SAC arm are not included, so nothing here speaks to the effect of an actively binding safety budget.

Seed 1

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	132.90	115.54	131.54	132.90	115.54	131.54	135.31	119.08	134.79
Mean episode cost	105.75 †	51.59	102.10	105.75	51.59	102.10	19.83	14.92	18.01
Violation — singularity	0.2279	0.1894	0.2242	0.2279	0.1894	0.2242	0.1867	0.1839	0.1804
Violation — joint limit	0.3632	0.1641	0.3539	0.3632	0.1641	0.3539	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0005	0.0000	0.0000	0.0005	0.0000	0.0000	0.0007	0.0000
Reward — lifting_object	9.461	9.125	9.443	9.461	9.125	9.443	9.486	8.952	9.475
Reward — reaching_object	0.900	0.812	0.897	0.900	0.812	0.897	0.913	0.820	0.912

Table A.1 — Training metrics for seed 1: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 2

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	134.02	117.44	133.90	134.02	117.44	133.90	135.54	115.99	134.92
Mean episode cost	7.86 †	8.83	7.65	7.86	8.83	7.65	17.25	16.16	16.65
Violation — singularity	0.1239	0.1250	0.1297	0.1239	0.1250	0.1297	0.2208	0.2259	0.2156
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0007	0.0000	0.0000	0.0007	0.0000	0.0000	0.0015	0.0000
Reward — lifting_object	9.503	9.146	9.494	9.503	9.146	9.494	9.493	8.821	9.484
Reward — reaching_object	0.912	0.806	0.911	0.912	0.806	0.911	0.912	0.794	0.911

Table A.2 — Training metrics for seed 2: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 3

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	133.77	117.77	133.07	133.77	117.77	133.07	131.81	113.97	132.57
Mean episode cost	163.06 †	107.84	162.30	163.06	107.84	162.30	12.86	10.11	11.93
Violation — singularity	0.9167	0.7069	0.9105	0.9167	0.7069	0.9105	0.1248	0.1255	0.1299
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0006	0.0000	0.0000	0.0006	0.0000	0.0000	0.0008	0.0000
Reward — lifting_object	9.514	9.158	9.492	9.514	9.158	9.492	9.416	8.705	9.430
Reward — reaching_object	0.907	0.811	0.905	0.907	0.811	0.905	0.899	0.782	0.900

Table A.3 — Training metrics for seed 3: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 4

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	134.53	120.67	134.29	134.53	120.67	134.29	132.58	112.11	131.50
Mean episode cost	31.58 †	22.95	30.04	31.58	22.95	30.04	20.02	13.78	19.69
Violation — singularity	0.3844	0.3210	0.3757	0.3844	0.3210	0.3757	0.1824	0.1473	0.1758
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0007	0.0000	0.0000	0.0007	0.0000	0.0000	0.0006	0.0000
Reward — lifting_object	9.438	9.147	9.478	9.438	9.147	9.478	9.446	8.811	9.433
Reward — reaching_object	0.907	0.827	0.909	0.907	0.827	0.909	0.896	0.762	0.887

Table A.4 — Training metrics for seed 4: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 5

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	133.74	115.79	132.74	133.74	115.79	132.74	135.33	120.44	135.05
Mean episode cost	18.88 †	15.84	19.09	18.88	15.84	19.09	22.53	21.26	24.15
Violation — singularity	0.2529	0.2187	0.2530	0.2529	0.2187	0.2530	0.4376	0.3790	0.4428
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0007	0.0000	0.0000	0.0007	0.0000	0.0000	0.0007	0.0000
Reward — lifting_object	9.472	9.117	9.424	9.472	9.117	9.424	9.491	8.937	9.464
Reward — reaching_object	0.906	0.815	0.900	0.906	0.815	0.900	0.914	0.823	0.913

Table A.5 — Training metrics for seed 5: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 50

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	132.57	118.16	132.14	132.57	118.16	132.14	131.63	112.70	132.02
Mean episode cost	8.95 †	7.43	8.60	8.95	7.43	8.60	25.00	19.62	23.85
Violation — singularity	0.1320	0.1102	0.1340	0.1320	0.1102	0.1340	0.3298	0.2610	0.3242
Violation — joint limit	0.0002	0.0001	0.0002	0.0002	0.0001	0.0002	0.0000	0.0000	0.0000
Violation — collision	0.0002	0.0009	0.0001	0.0002	0.0009	0.0001	0.0000	0.0012	0.0000
Reward — lifting_object	9.430	9.087	9.421	9.430	9.087	9.421	9.423	8.793	9.428
Reward — reaching_object	0.901	0.814	0.902	0.901	0.814	0.902	0.898	0.781	0.898

Table A.6 — Training metrics for seed 50: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 51

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	137.40	115.60	137.28	137.40	115.60	137.28	133.46	114.30	132.98
Mean episode cost	2.02 †	10.22	1.78	2.02	10.22	1.78	16.64	14.87	17.02
Violation — singularity	0.0825	0.1378	0.0784	0.0825	0.1378	0.0784	0.2607	0.2349	0.2618
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0008	0.0000	0.0000	0.0008	0.0000	0.0000	0.0010	0.0000
Reward — lifting_object	9.512	9.091	9.510	9.512	9.091	9.510	9.420	8.897	9.424
Reward — reaching_object	0.919	0.819	0.918	0.919	0.819	0.918	0.903	0.786	0.902

Table A.7 — Training metrics for seed 51: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 52

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	132.42	118.24	132.08	132.42	118.24	132.08	133.74	111.06	132.72
Mean episode cost	104.72 †	69.33	106.93	104.72	69.33	106.93	9.43	7.37	9.50
Violation — singularity	0.6733	0.5329	0.6808	0.6733	0.5329	0.6808	0.2302	0.1683	0.2324
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0005	0.0000	0.0000	0.0005	0.0000	0.0000	0.0013	0.0000
Reward — lifting_object	9.445	9.121	9.439	9.445	9.121	9.439	9.443	8.683	9.458
Reward — reaching_object	0.905	0.809	0.903	0.905	0.809	0.903	0.897	0.755	0.891

Table A.8 — Training metrics for seed 52: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 53

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	135.07	119.94	134.34	135.07	119.94	134.34	129.42	113.07	130.63
Mean episode cost	18.84 †	14.88	18.81	18.84	14.88	18.81	25.73	20.30	23.51
Violation — singularity	0.3095	0.2554	0.3083	0.3095	0.2554	0.3083	0.5453	0.4143	0.5166
Violation — joint limit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Violation — collision	0.0000	0.0005	0.0000	0.0000	0.0005	0.0000	0.0000	0.0005	0.0000
Reward — lifting_object	9.483	9.124	9.473	9.483	9.124	9.473	9.395	8.854	9.430
Reward — reaching_object	0.909	0.825	0.908	0.909	0.825	0.908	0.883	0.790	0.887

Table A.9 — Training metrics for seed 53: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.

Seed 54

Metric	PPO			ctrl			cPPO		
	final	mean	tail	final	mean	tail	final	mean	tail
Mean reward	135.15	112.49	134.34	135.15	112.49	134.34	131.34	107.48	129.34
Mean episode cost	10.86 †	13.28	7.95	10.86	13.28	7.95	12.56	10.05	11.98
Violation — singularity	0.0793	0.1322	0.0842	0.0793	0.1322	0.0842	0.1223	0.1100	0.1265
Violation — joint limit	0.0432	0.0047	0.0276	0.0432	0.0047	0.0276	0.0000	0.0000	0.0000
Violation — collision	0.0002	0.0007	0.0001	0.0002	0.0007	0.0001	0.0002	0.0013	0.0001
Reward — lifting_object	9.516	9.137	9.481	9.516	9.137	9.481	9.425	8.630	9.387
Reward — reaching_object	0.911	0.798	0.908	0.911	0.798	0.908	0.882	0.738	0.865

Table A.10 — Training metrics for seed 54: PPO, ctrl and cPPO (budget 25). Values are final / full-run mean / final-10% tail mean over 1500 iterations.